

Measuring ROI on Sponsored Search Ads at Bazaar.com

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Executive Summary

Background - Bazaar.com is a leading U.S. online retailer that runs sponsored search advertising on multiple search engines, including Google, Bing, Yahoo, and Ask. A meaningful portion of this spend supports branded keyword campaigns — search terms that include the Bazaar brand name (e.g., “Bazaar shoes”). The marketing team currently reports a 320% ROI on Google branded sponsored ads, calculated by attributing all sponsored-click revenue to the ads themselves.

Business question - Are these branded sponsored ads truly generating incremental revenue, or are they paying for clicks from customers who would have visited Bazaar.com anyway through the free organic search link? In other words, is the reported 320% ROI a real measure of value, or is it inflated by clicks the company would have received for free?

Natural experiment and method - A technical glitch turned off Google sponsored ads in weeks 10–12 of a 12-week observation window, while sponsored ads on Bing, Yahoo, and Ask continued running normally. This unintended interruption gave us a way to measure what actually happens when the ads disappear: we compare Google’s traffic change with the change observed on the other three platforms over the same period, which lets us separate the effect of the ads from broader market trends affecting all search engines.

Key findings. A simple before-vs-after comparison on Google alone suggest the ads barely mattered (a small, statistically insignificant decline of 1,846 clicks/week). This estimate is misleading because it ignores the fact that the broader search market grew substantially during weeks 10–12 — visible in the other platforms’ traffic. Once we account for that market-wide growth, turning off Google ads reduced total weekly traffic by approximately **9,911 clicks/week** beyond the trend on the other platforms (significant at $p < 0.01$). The traffic loss is consistent with cannibalization: Google lost ~12,204 sponsored clicks/week but recovered ~2,293 organic clicks/week, meaning roughly **19% of sponsored clicks would have arrived organically anyway**. Applying the corrected incremental traffic to Bazaar’s per-click economics yields a corrected campaign-level ROI of approximately **580%** — Bob’s 320% figure is wrong, but the corrected estimate is still positive.

Recommendation. Continue running the Google branded sponsored search campaign. Bob’s 320% calculation should not be used going forward. Validate the corrected estimate with a controlled geographic experiment before scaling spend, and recognize that this conclusion applies only to branded keywords.

1 What is wrong with Bob’s ROI calculation?

Bob’s logic uses cost per sponsored click (\$0.60), conversion probability (12%), and margin per conversion (\$21) to compute a revenue per click of \$2.52, giving an ROI of $(2.52 - 0.60) / 0.60 = 320\%$.

The problem is that this assumes *every* sponsored click is a new visit caused by the ad. For branded keywords, this assumption is implausible: a user typing “Bazaar shoes” already knows about Bazaar. If the sponsored ad is absent, that user typically clicks the Bazaar organic listing displayed immediately below — producing the same purchase with no ad spend. The sponsored ad therefore *cannibalizes* a click that organic search would have delivered for free.

The descriptive evidence supports this. Comparing average weekly Google traffic before and during the glitch:

Period	Sponsored	Organic	Total
Weeks 1–9 (Google)	31,390	125,195	156,585
Weeks 10–12 (Google)	0	148,477	148,477
Difference	-31,390	+23,282	-8,108

When Google ads stopped, organic clicks rose by ~23,282/week - those users substituted to the organic link rather than disappearing. Only ~8,108 clicks/week were genuinely lost from total traffic. The true value of a sponsored click is the revenue from clicks that *would not* have occurred without the ad.

Disclaimer - This table contains numbers from the case study file and not the csv shared on Canvas.

2 Treatment, Control, and the Natural Experiment

Unit of observation: The unit of observation in this data set is a search engine platform in a given week - in other words, each row of the data represents one platform observed at one point in time. With 4 platforms and 12 weeks, we have 48 observations

Treatment: the cessation of sponsored ads on Google starting in week 10. Because the shutdown was caused by a technical glitch (not a strategic choice), the treatment is plausibly exogenous — the defining feature of a natural experiment.

Treated unit: Google. **Control units:** Yahoo, Bing, Ask — sponsored ads continued running on these platforms throughout all 12 weeks.

Pre-period: weeks 1–9. **Post-period:** weeks 10–12.

Outcome variable: `total_traffic = avg_spons + avg_org`. We use total traffic (not just sponsored or just organic) because the business question is whether ads bring incremental visitors to Bazaar.com. If lost sponsored clicks are fully replaced by organic, total is unchanged and the ads added no value; if total falls, the ads were generating incremental visits.

3 First-Difference (pre-post) Estimate

We first estimate the simple pre-post change using only Google data, regressing total traffic on a post indicator.

```
# Restrict to Google only; the coefficient on `post` is the pre-post difference
google_only <- data %>% filter(platform == "goog")
prepost_levels <- lm(total_traffic ~ post, data = google_only)
summary(prepost_levels)
```

```
##
## Call:
## lm(formula = total_traffic ~ post, data = google_only)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -7003.9 -2630.1  -172.5   2088.4   8625.1
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)      8390         1598   5.252 0.000373 ***
## post           -1846         3195  -0.578 0.576238
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 4793 on 10 degrees of freedom
## Multiple R-squared:  0.0323, Adjusted R-squared:  -0.06447
## F-statistic: 0.3337 on 1 and 10 DF,  p-value: 0.5762
```

Interpretation.

- **Intercept = 8,390:** Google's average weekly traffic in the pre-period (weeks 1–9), when ads were running.
- **post coefficient = -1,846:** Google's average traffic dropped by ~1,846 clicks/week in the post-period, giving an average of $8,390 - 1,846 = 6,544$ clicks/week in weeks 10–12.
- **Standard error = 3,195, p = 0.576:** the estimate is *not* statistically significant.

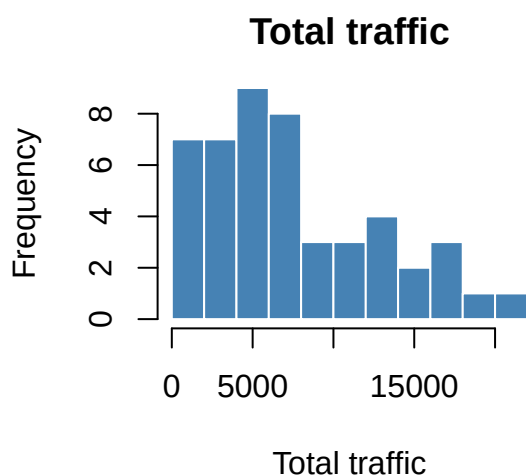
Why we cannot rely on this estimate. Taken at face value, the pre-post estimator says ads caused only a small, insignificant drop — which would imply ads are doing little. This is misleading. The pre-post estimator confounds the treatment effect with anything else that changed between weeks 1–9 and weeks 10–12 (seasonality, market growth, holidays, search demand surges). The control platforms reveal that the broader search market was growing strongly during weeks 10–12, meaning Google should have been growing too. A flat-to-slightly-declining Google trajectory actually represents a substantial loss relative to its proper counterfactual. The DiD estimator, which we turn to next, captures this missed growth.

4 Difference-in-differences Estimate

4.1 Skewness check

Total traffic is right-skewed and grows multiplicatively across weeks. We log-transform it for the DiD specification — this both reduces skewness and makes the parallel-trends assumption more defensible on the proportional scale.

```
par(mfrow = c(1, 2), mar = c(4, 4, 2.5, 1))
hist(data$total_traffic, breaks = 12, col = "steelblue", border = "white",
     main = "Total traffic", xlab = "Total traffic")
```



We estimate the DiD using the log of total traffic due to the skewness.

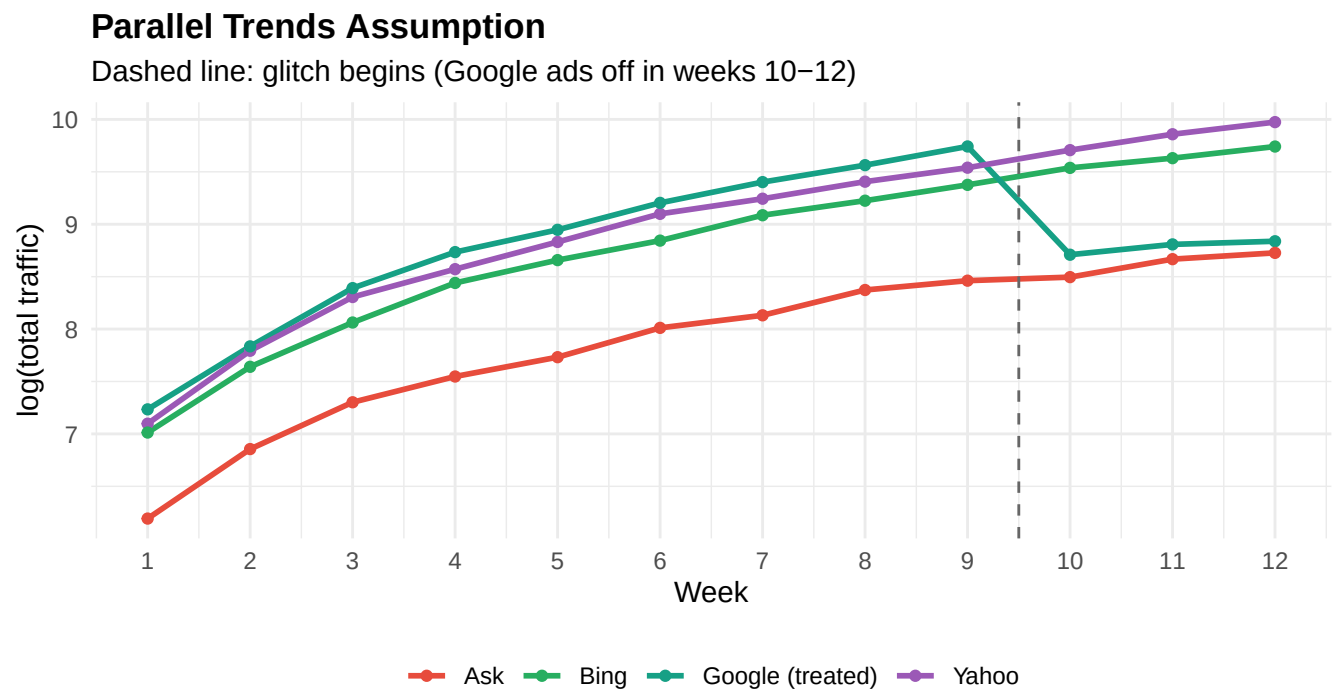
4.1.1 Assumptions required for causal inference

Two assumptions must hold for the DiD estimate to be interpreted causally:

1. **Parallel trends.** In the absence of the glitch, Google's traffic would have continued moving in parallel with Bing, Yahoo, and Ask. We cannot test this directly for the post-period (it is counterfactual), but we can assess it visually in the pre-period — the four platforms move together in weeks 1–9 (see plot below), supporting the assumption.
2. **No interference (SUTVA).** The treatment of one platform must not affect outcomes on the other platforms. In our setting, the glitch on Google shouldn't change behavior on Bing, Yahoo, or Ask. This is plausible because each platform's users searched independently — but if some Google users switched to Bing because Bazaar's ad was missing, the controls would be contaminated and the DiD would be biased.

4.2 Parallel Trends Assumption

```
ggplot(data, aes(x = week, y = log_traffic, color = platform)) +  
  geom_line(linewidth = 1) +  
  geom_point(size = 1.5) +  
  geom_vline(xintercept = 9.5, linetype = "dashed", color = "gray40") +  
  scale_x_continuous(breaks = 1:12) +  
  scale_color_manual(  
    name = NULL,  
    values = c("ask" = "#E74C3C",  
               "bing" = "#27AE60",  
               "goog" = "#16A085",  
               "yahoo" = "#9B59B6"),  
    labels = c("ask" = "Ask",  
               "bing" = "Bing",  
               "goog" = "Google (treated)",  
               "yahoo" = "Yahoo")  
  ) +  
  labs(title = "Parallel Trends Assumption",  
       subtitle = "Dashed line: glitch begins (Google ads off in weeks 10-12)",  
       x = "Week", y = "log(total traffic)") +  
  theme_minimal(base_size = 11) +  
  theme(legend.position = "bottom",  
        plot.title = element_text(face = "bold"))
```



In weeks 1–9 all four platforms trend upward together, supporting parallel trends. After week 9, Google diverges sharply downward while the controls continue rising — exactly the pattern a real treatment effect would produce.

4.3 DiD on Log Traffic Scale

```
# DiD on log scale
FE_did <- plm(log_traffic ~ post + treat_post,
              data = data,
              index = "platform",
              effect = "individual",
              model = "within")
summary(FE_did)

## Oneway (individual) effect Within Model
##
## Call:
## plm(formula = log_traffic ~ post + treat_post, data = data, effect = "individual",
##      model = "within", index = "platform")
##
## Balanced Panel: n = 4, T = 12, N = 48
##
## Residuals:
##      Min.      1st Qu.      Median      3rd Qu.      Max.
## -1.575180 -0.236438  0.037839  0.453689  0.958344
##
## Coefficients:
##              Estimate Std. Error t-value Pr(>|t|)
## post           1.11764    0.26744  4.1791 0.000145 ***
## treat_post     -1.11634    0.53487 -2.0871 0.042985 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Total Sum of Squares:    28.708
## Residual Sum of Squares: 20.276
## R-Squared:              0.2937
## Adj. R-Squared: 0.20962
## F-statistic: 8.73248 on 2 and 42 DF, p-value: 0.00067414
```

Interpretation. - **post = 1.118 (p < 0.001)**: the time trend across the control platforms. In level terms, $\exp(1.118) - 1$ is approximately +206% — the controls roughly tripled their traffic from the pre- to post-period. A large market-wide surge was occurring. - **treat_post = -1.116 (p = 0.043)**: the DiD estimate. In level terms, $\exp(-1.116) - 1$ is approximately -67%: turning off Google ads caused total traffic to be about 67% lower than it would have been if ads had continued, holding the control trend fixed.

Comparison with the first-difference estimate. The pre-post estimate (-1,846 clicks/week, not significant) suggested ads barely mattered. The DiD reveals the opposite: ads were generating substantial incremental traffic, but the simple pre-post estimate was masked by a large background market trend. This is exactly why a control group is essential — without it, we mistakenly attributed the absence of expected growth to “no effect” when it was actually a large missed opportunity.

4.4 DiD for ROI calculation

The log-scale DiD gives a percentage interpretation. For the dollar-based ROI calculation in Q5, we re-run the DiD on raw levels so that the coefficient is interpretable as clicks/week.

```
# DiD on level scale; treat_post coefficient = lost clicks per week from turning ads off
did_levels <- lm(total_traffic ~ treated + post + treat_post, data = data)
summary(did_levels)
```

```
##
## Call:
## lm(formula = total_traffic ~ treated + post + treat_post, data = data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -8437.7 -3231.0  -510.5   3591.6   8630.0
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    5265.0      882.5   5.966 3.79e-07 ***
## treated         3124.9     1765.0   1.770  0.08357 .
## post           8064.7     1765.0   4.569 3.94e-05 ***
## treat_post     -9910.6     3530.0  -2.808  0.00741 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 4586 on 44 degrees of freedom
## Multiple R-squared:  0.3274, Adjusted R-squared:  0.2816
## F-statistic: 7.141 on 3 and 44 DF, p-value: 0.0005211
```

Interpretation of the DiD coefficients:

Coefficient	Estimate	Meaning
Intercept	5,265	Average control traffic in pre-period (weeks 1–9)
treated	+3,125	Baseline gap: Google’s pre-period traffic exceeds the controls’

Coefficient	Estimate	Meaning
post	+8,065	Time trend: controls grew by ~8,065 clicks/week from pre to post
treat_post	-9,911	DiD: extra change for Google beyond the control trend

If Google had ridden the same wave as the controls, it should have grown by +8,065 clicks/week (from 8,390 to 16,455). Instead it shrank by 1,846 (down to 6,544). The 9,911-click gap is the true incremental traffic the ads were generating each week, statistically significant at $p < 0.01$.

5 Corrected ROI calculation

5.1 Decomposing the Effect: sponsored vs. organic substitution

```
# Decompose the total effect into sponsored loss and organic substitution
did_spons <- lm(avg_spons ~ treated + post + treat_post, data = data)
did_org <- lm(avg_org ~ treated + post + treat_post, data = data)
summary(did_spons)$coefficients["treat_post", ]
```

```
##      Estimate      Std. Error      t value      Pr(>|t|)
## -1.220381e+04  2.679047e+03 -4.555283e+00  4.123484e-05
```

```
summary(did_org)$coefficients["treat_post", ]
```

```
##      Estimate      Std. Error      t value      Pr(>|t|)
##  2.293222e+03  8.614328e+02  2.662102e+00  1.080272e-02
```

Outcome	treat_post coefficient	Interpretation
Sponsored clicks	-12,204	Sponsored clicks lost when ads turned off
Organic clicks	+2,293	Organic clicks gained (cannibalization recovery)
Total traffic	-9,911	Net loss = -12,204 + 2,293

5.2 Computing the Corrected ROI


```

# Average sponsored clicks per week for Google during the pre-glitch period
# This is what Bazaar was actually paying for in a "normal" ads-on week
avg_spons_goog_pre <- data %>%
  filter(platform == "goog", week <= 9) %>%
  summarise(mean(avg_spons)) %>%
  pull()
avg_spons_goog_pre

```

```
## [1] 6123.222
```

```

# Inputs to the corrected ROI calculation
incremental_clicks <- 9911 # from level-scale DiD
cost_per_click <- 0.60
conv_rate <- 0.12
margin_per_conv <- 21

cost_per_week <- avg_spons_goog_pre * cost_per_click
revenue_per_week <- incremental_clicks * conv_rate * margin_per_conv
roi_corrected <- (revenue_per_week - cost_per_week) / cost_per_week

cost_per_week

```

```
## [1] 3673.933
```

```
revenue_per_week
```

```
## [1] 24975.72
```

```
roi_corrected
```

```
## [1] 5.798087
```

The reasoning.

- **Incremental clicks per week = 9,911.** This is the absolute value of the DiD coefficient on total traffic – the truly net-new visits the ads were generating. (Equivalently: the visits Bazaar would lose if it stopped buying ads.)
- **Cost per week is approximately \$3,674.** Bazaar pays \$0.60 for *every* sponsored click, including the cannibalized ones. We use the pre-glitch average of ~6,123 sponsored clicks/week to represent a typical ads-on week.
- **Revenue per week is approximately \$24,976.** Apply Bazaar's conversion rate (12%) and margin (\$21) to the 9,911 incremental visits.
- **Corrected ROI = (\$24,976 - \$3,674) / \$3,674, which is approximately 580%.**

5.3 Caveats and Recommendation

5.3.1 Caveats

- The post-period contains only 3 weeks, giving wide confidence intervals on all estimates. The DiD coefficient is statistically significant ($p < 0.01$) but the magnitude carries meaningful uncertainty.
- The control surge in weeks 10–12 is large. If Google was systematically less able to capture that surge for platform-specific reasons unrelated to ads, the parallel-trends assumption could be overstating the effect.
- Results apply only to **branded keywords**. Non-branded keywords (“shoes” rather than “Bazaar shoes”) have no organic Bazaar listing to substitute for the ad, so cannibalization is much weaker and incremental ROI is likely higher there.

5.3.2 Recommendation

1. **Recalculate the ROI** Bob’s methodology is biased because it counts cannibalized clicks as incremental. The corrected ROI is approximately 580%.
2. **Continue the branded sponsored search campaign on Google.** Even under the conservative ROI framing, the incremental return justifies the spend.
3. **Validate before scaling.** Run a controlled geographic A/B test (turn off ads in select branded campaigns in matched cities for 4–6 weeks) to confirm the magnitude. The natural experiment was useful but not designed — purpose-built experiments will give tighter estimates.
4. **Do NOT generalize to non-branded campaigns.** The cannibalization mechanism is specific to branded searches, where an organic Bazaar listing exists.
5. **Consider re-evaluating bid strategy on branded keywords.** Since ~19% of paid clicks are cannibalized from organic, lowering bids on the most competitive branded terms (where Bazaar’s organic listing already ranks #1) could reduce wasted spend without losing meaningful traffic.